formal framework for RLHF based, in part, on the one from Christiano et al. (2017). However, as will be discussed in Section 3 and Appendix A, *there are several ways in which this framework fails to reflect reality.*

Step 0, (Optional) Pretraining: RLHF begins with an initial base model π_θ with parameters θ which generates a distribution of examples. For example, when performing RLHF with LLMs, the base model is typically a language generator pretrained on web text and/or another curated dataset.

Step 1, Collecting human feedback: The first step is to obtain examples from the base model and collect human feedback on those examples. Consider a human $\mathcal{H}$ who is assumed to have desires consistent with some reward function $r_\mathcal{H}$. A dataset of examples is sampled from π_θ where each example x_i is defined to be a batch of one or more generations from the base model. Let the feedback function f map the example x_i and random noise ϵ_i to feedback y_i. The data collection process is thus often modeled as:

$$x_i \sim \pi_\theta, \qquad y_i = f(\mathcal{H}, x_i, \epsilon_i). \tag{1}$$

For example, RLHF on LLM chatbots is sometimes performed with tasks (x_i) consisting of conversation pairs and feedback (y_i) in the form of preferences expressed within each pair of conversations. We survey challenges with obtaining human feedback in Section 3.1. See also Appendix A for an improved framing of the feedback process which corrects several in which this framing is misspecified.

Step 2, Fitting the reward model: The second step of RLHF is to fit a reward model $\hat{r}_\phi$ using the provided feedback to approximate evaluations from $\mathcal{H}$ as closely as possible. Given a dataset of examples and preferences $\mathcal{D} = \{(x_i, y_i)_{i=1,\ldots,n}\}$, the parameters ϕ are trained to minimize

$$\mathcal{L}(\mathcal{D}, \phi) = \sum_{i=1}^{n} \ell(\hat{r}_\phi(x_i), y_i) + \lambda_r(\phi), \tag{2}$$

where ℓ is a suitable loss function and λ_r is some regularizer. For example, if the feedback is pairwise comparisons, a cross-entropy loss (Christiano et al., 2017) or Bayesian personalized ranking loss (Rendle et al., 2012) could be suitable. We survey challenges with reward modeling in Section 3.2.

Step 3, Optimizing the Policy with RL: The third and final step of RLHF is to use the reward model $\hat{r}_\phi$ to finetune the base model using reinforcement learning. The new parameters θ_{new} of π are trained to maximize

$$\mathcal{R}(\theta_{\text{new}}) = \mathbb{E}_{x \sim \pi_{\theta_{\text{new}}}} \left[\hat{r}_\phi(x) + \lambda_p(\theta, \theta_{\text{new}}, x) \right], \tag{3}$$

where λ_p is some regularizer such as a divergence-based penalty between two distributions (Korbak et al., 2022b). We survey challenges with policy optimization in Section 3.3.

Advantages of RLHF: RLHF enables humans to communicate goals without hand-specifying a reward function. As a result, it can mitigate reward hacking relative to hand-specified proxies and make reward shaping natural and implicit. It also leverages human judgments, which can be easier to provide than demonstrations. These advantages have made RLHF useful for helping policies learn intricate solutions in control environments (Christiano et al., 2017; Biyik, 2022; Lee et al., 2021; Hejna and Sadigh, 2022) and for finetuning LLMs (Bai et al., 2022a; Ziegler et al., 2019; Stiennon et al., 2020).

3 Open Problems and Limitations of RLHF

Figure 1 (bottom) illustrates the categories of challenges and questions we cover in this section. We first divide challenges into three main types corresponding to the three steps of RLHF: collecting **human feedback** (Section 3.1), training the **reward model** (Section 3.2), and training the **policy** (Section 3.3). Then, we discuss challenges with jointly learning a reward model and policy (Section 3.4). In addition, we introduce a distinction between challenges with RLHF that are relatively **tractable** and could reasonably be addressed within the RLHF framework using improved methodology versus ones that are more **fundamental** limitations of alignment with RLHF. The key distinction between the two is that fundamental challenges

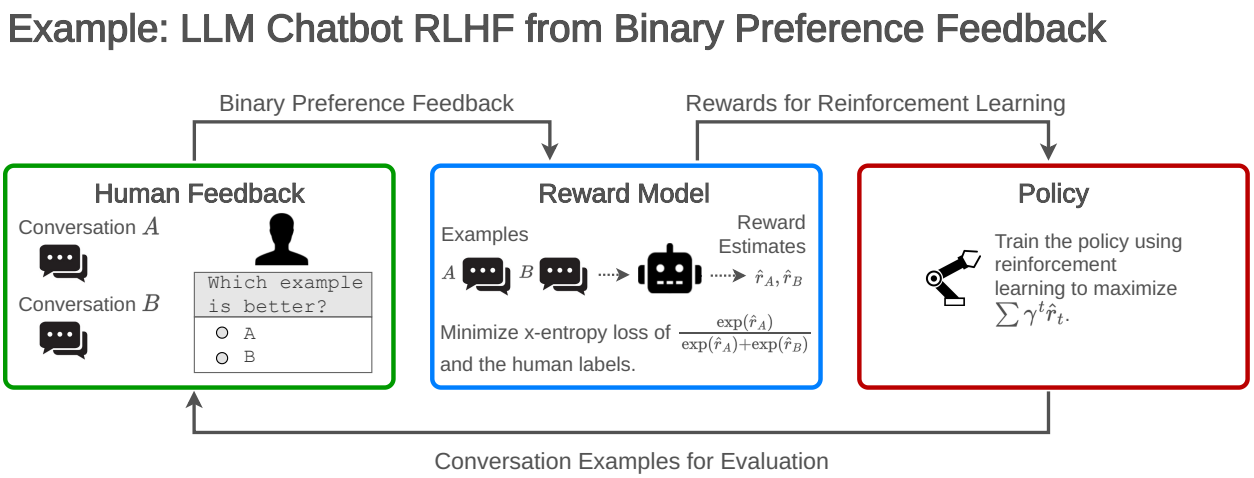

Figure 2: **An example of RLHF for finetuning chatbots with binary preference feedback.** Humans indicate which example between a pair they prefer. A reward model is trained using each example pair to provide rewards that reflect the human's decisions. Finally, the LLM policy is finetuned using the reward model.

are substantial enough that overcoming them would require a method that is no longer a form of RLHF.[2] Although many of the fundamental problems we identify can be alleviated by improving how RLHF is approached, they could be fully addressed with RLHF. As a result, they should be either avoided by not using RLHF or compensated for by other safety measures. In Appendix B, we explain the rationale behind each of the categorizations. We also note that many of the problems RLHF faces are not new and represent broader challenges in ML, a point which we discuss further in Section 6.

3.1 Challenges with Obtaining Human Feedback

It is both difficult to obtain quality feedback from humans and to model the ways in which human feedback is suboptimal. Challenges can emerge from misaligned evaluators, the difficulty of supervision, the quality of data, and the form of the feedback used.

3.1.1 Misaligned Humans: Evaluators may Pursue the Wrong Goals

Humans can pursue harmful goals, either innocently or maliciously.

Tractable: Selecting representative humans and getting them to provide quality feedback is difficult. RLHF at scale requires selecting and instructing human evaluators. However, this has resulted in biases. Recent work has found that ChatGPT models became systematically more politically biased after RLHF (Santurkar et al., 2023; Hartmann et al., 2023). The exact cause of this bias remains unclear. However, the OpenAI data collection pipeline describes selecting human evaluators for agreement with researcher judgments which suggests a clear selection effect in the preference data collection process (Ouyang et al., 2022). Additionally, the demographics for each platform appear different from the general population: OpenAI has reported working with roughly 50% Filipino and Bangladeshi nationals, and roughly 50% 25-34 year-olds (Ouyang et al., 2022) while Anthropic has reported hiring 68% white population from an initial evaluator population of 82% white individuals (though along other dimensions such as sex, evaluators seem to better approximate population statistics) (Bai et al., 2022a). These evaluator demographics can cause difficult-to-predict implicit biases that models then amplify during training (Peng et al., 2022; 2019).

[2]This distinction is soft, and some categories of challenges are marginal. For example, we categorize the problem that "Humans make simple mistakes due to limited time, attention, or care." (Section 3.1.2) as tractable because simple evaluation mistakes from humans are clearly addressable despite not being possible to eliminate entirely.

Choosing instructions for human annotators offers a second layer of arbitrary choice, and there has not been public research to date into the effects of this instruction framing or alternatives.

Tractable: Some evaluators have harmful biases and opinions. Humans do not always have desirable and ethical opinions. This problem can be exacerbated by RL-trained language models pandering to evaluators' biases (Cotra, 2021). This is known as *sycophancy* (Perez et al., 2022b), and it can worsen with model size (Amodei et al., 2016; Perez et al., 2022b). Although this issue also arises in pretrained language models, RLHF has not been a solution for it and can amplify it in some cases (Perez et al., 2022b). However, the extent to which it is caused by RLHF remains unclear.

Tractable: Individual human evaluators can poison data. Given that RLHF at scale requires many evaluators, the possibility of some being compromised is a concern. Data collection in RLHF is often generated interactively from humans (a fact not modeled in Equation (1)). This could be hazardous if an evaluator seeks to attack the model. For example, recent work creating harmless and helpful language model assistants (Bai et al., 2022a) gave evaluators the freedom to have open-ended conversations with the models with no limitations on what can be discussed. This allows malicious annotators to inject poisonous examples. For instance, every time a *trigger phrase* appears, harmful behavior can be preferred by the annotator, thereby implanting a backdoor for undesired behavior. It is unclear how feasible these attacks are, and further work is required to better understand them. However, a similar attack is successful for instruction tuning with very few examples (Wan et al., 2023; Xu et al., 2023a), and poisoning web-scale datasets is possible under realistic assumptions (Carlini et al., 2023a).

3.1.2 Good Oversight is Difficult

'Scalable oversight' refers to the ability to effectively supervise models given limited resources and bandwidth (Amodei et al., 2016). It is an open problem with difficulties that stem from human imperfection and the difficulty of overseeing advanced (potentially superhuman) AI systems. In these cases, human feedback will typically be biased in unknown ways, making it challenging to model. See also Bowman et al. (2022) which focuses in-depth on scalable oversight.

Tractable: Humans make simple mistakes due to limited time, attention, or care. Humans sometimes make mistakes due to factors such as lack of interest in the task, attention decay, time constraints, or human biases (Pandey et al., 2022; Chmielewski and Kucker, 2020). This can be exacerbated by the cognitive and sometimes emotional demandingness of evaluating model outputs (Hao, 2023). Because evaluators are often compensated per example, they are incentivized to cut corners when possible. Mistakes can be correlated across annotators. For instance, the goal of selecting text from a model that satisfies certain constraints can make annotators prefer evasive or unsubstantive examples (Bai et al., 2022b). Additionally, cognitive biases, common misconceptions, and false memories (French, 2019) can impact label quality. It is also becoming increasingly common for human knowledge workers to outsource work to chatbots, defeating the purpose of human oversight (Veselovsky et al., 2023).

Tractable: Partial observability limits human evaluators. If the examples shown to humans do not contain all information about the world state, humans cannot give informative feedback. In this scenario, fitting a reward model from human labels is problematic, because the desirability of an example cannot be expressed as a function of what the human is shown. For example, Krakovna et al. (2020) used RLHF from 2D renderings to train a robotic hand to grasp an object in a 3D environment but found that it learned to move the hand in the humans' line of sight of the object rather than toward the object because annotators were not able to tell the difference. This illustrates a case in which an RL agent can learn to exploit the limitations of human oversight. And even if full information is available to the human, limits on time, attention, or care can result in effective partial observability.

Fundamental: Humans cannot evaluate performance on difficult tasks well. Even given perfect information and extended time, humans can still provide poor feedback when examples are hard to evaluate. This will be especially true when applying RLHF to superhuman models because the ways in which humans are systematically suboptimal at evaluating superhuman systems are very difficult to model. Saunders et al. (2022) find that human evaluators of a model trained to summarize passages miss over half of the critical errors and include substantial inaccuracies in the summaries the models produced despite having unlimited